University of Science and Technology of Hanoi

Group 1 - ICT Department

Natural language Processing Course Report

# Vietnamese Hate Speech Detection using PhoBERT and SVM

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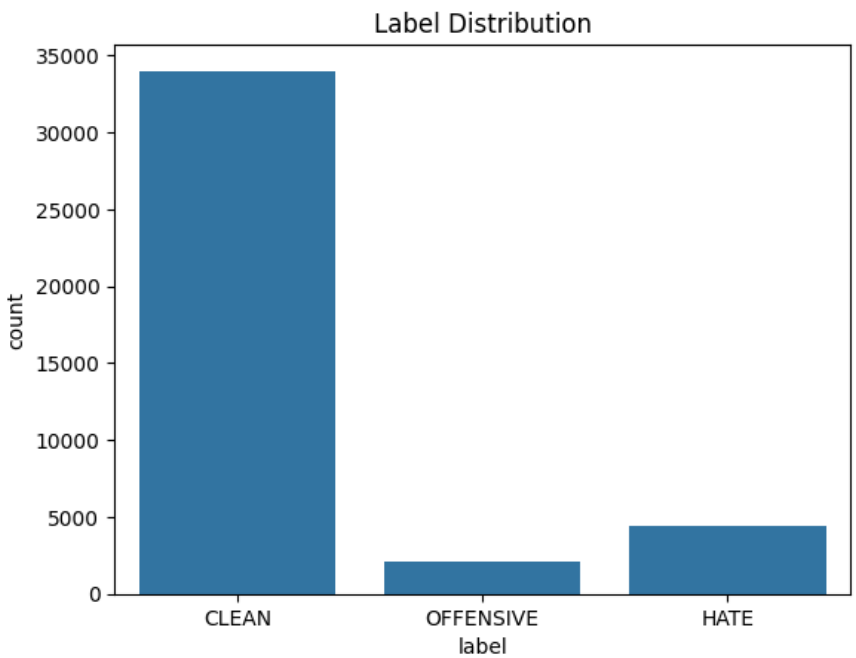
# 1 Project Overview

The goal of this project is to build an automated system to classify Vietnamese social media comments into three categories: `CLEAN`, `OFFENSIVE`, and `HATE`. This report details the comparative analysis between a traditional machine learning baseline (TF-IDF + SVM) and a deep learning approach using the PhoBERT transformer model.

# 2 Data Analysis

The dataset utilized is the `ViHSD` (Vietnamese `HATE` Speech Detection) dataset. Initial exploratory data analysis reveals a significant class imbalance.




Hình 1: Label Distribution: High prevalence of `CLEAN` samples compared to `OFFENSIVE` and `HATE`.

As shown in Figure 1, the `CLEAN` class dominates the dataset, which suggests that **Macro F1-score** is a more critical evaluation metric than simple accuracy to ensure the model performs well on the minority classes.

# 3 Methodology

Building upon the preprocessed data and exploratory analysis presented in the previous section, this stage focuses on converting the cleaned text into machine-readable formats and implementing the classification architectures.

## 3.1 Text Representation

To process the raw Vietnamese text, the models require specific numerical representations. For the statistical baseline, a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer was

employed using an n-gram range of (1, 2), capturing both individual words and two-word sequences and weighting them by their importance within a document relative to the entire corpus. To keep the feature space manageable under computational constraints, the vocabulary was capped at 20,000 features, meaning less frequent unigrams and bigrams are discarded during vectorization. For the deep learning approach, text was processed using the `vinai/phobert-base` tokenizer, which segments text into sub-word units and applies truncation to a fixed length of 128 tokens to ensure all input tensors maintain uniform dimensions for the neural network.

### 3.2 Baseline Model: Linear SVM

A Support Vector Machine (SVM) utilizing a linear kernel (`LinearSVC`) was selected as the baseline classifier. The TF-IDF vectorizer with a (1, 2) n-gram range produces a high-dimensional feature space, as each unique unigram and bigram in the corpus becomes its own dimension. In such high-dimensional spaces, text classes tend to become linearly separable, making a linear kernel a natural and well-justified choice. Furthermore, the project was conducted under severe computational limitations, which directly contributed to the decision to use a linear kernel over more resource-intensive non-linear alternatives. This approach provides an efficient computational profile while establishing a robust performance benchmark for the transformer-based model.

### 3.3 Advanced Model: PhoBERT Fine-Tuning

To contrast the statistical baseline, the project utilizes the `vinai/phobert-base` architecture, a language model pre-trained specifically on large-scale Vietnamese datasets. The model was fine-tuned for 10 epochs using the HuggingFace `Trainer` API with a learning rate of 2e-5, a batch size of 64, and a weight decay of 0.01 for regularization. Mixed precision training (`fp16`) was also enabled to reduce memory consumption and accelerate training — a practical necessity given the computational constraints of the project. By leveraging the self-attention mechanism of the transformer architecture, the model can potentially capture semantic context and nuances in Vietnamese slang or offensive metaphors more effectively than traditional statistical models. Following the attention layers, the final hidden state is fed into a classification head that produces logits for the three target classes, with the final prediction computed by taking the argmax of the logits.

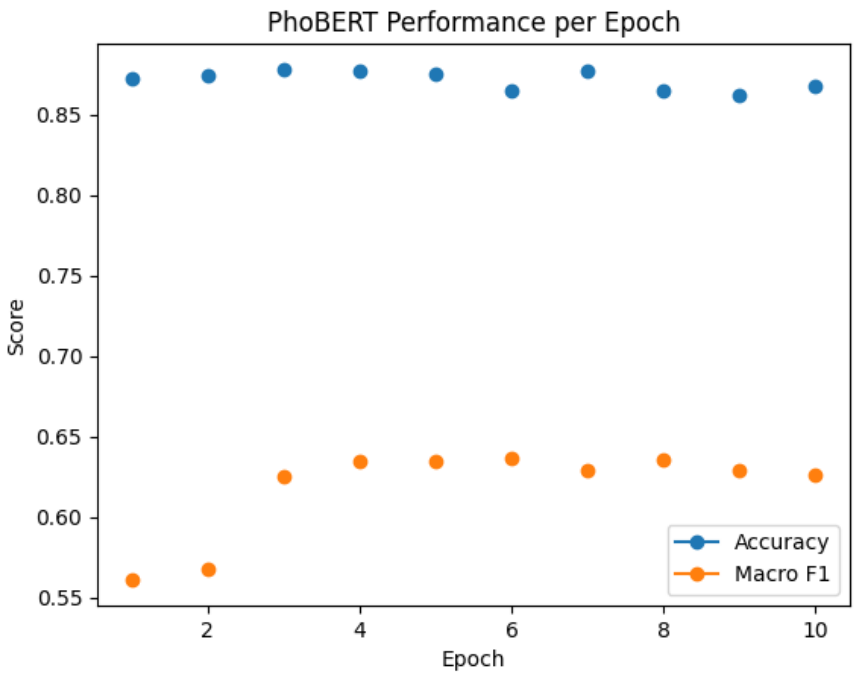
### 3.4 Interactive System Deployment

To bridge the gap between research and practical application, a real-time demonstration environment was established using Streamlit and Pyngrok. A web interface was developed with Streamlit, providing a simple text input alongside a model selector that allows users to switch between PhoBERT and TF-IDF + SVM and compare their predictions directly. Since the server runs locally within the Colab environment, Pyngrok is used to create a secure tunnel that exposes the local Streamlit server (running on port 8501) as a publicly accessible URL, eliminating the need for any local environment setup on the user's end.

The demo can be self-hosted by anyone with access to the project files. The process involves uploading the notebook to Google Colab, selecting a T4 GPU instance under the runtime settings, and uploading the required checkpoint files — `phobert_model.zip` for the transformer and `svm_model.pkl` together with `tfidf_vectorizer.pkl` for the baseline. Running the notebook end-to-end will then launch the Streamlit interface and print a public Pyngrok URL that can be shared and accessed from any browser.

# 4 Evaluation and Results

## 4.1 PhoBERT Training Performance




Hình 2: PhoBERT Model Performance after 10 epochs

The PhoBERT model was fine-tuned over 10 epochs. This extended training duration allowed the model to reach convergence. While the overall accuracy remained consistent (approximately 0.87) from the first epoch, the Macro F1 score demonstrated a clear learning curve. The F1 score rose from 0.56 in epoch 1 before stabilizing around 0.62 to 0.63 by epoch 4. This early plateau, which was maintained through the 10th epoch, indicates that the model converged and reached its learning capacity for this specific configuration and dataset.

## 4.2 Comparative Analysis: Baseline vs. Deep Learning

Because the dataset is heavily imbalanced toward the `CLEAN` class, overall accuracy is a misleading metric. The **Macro F1-score** and **Macro Recall** are the most critical indicators of performance for the minority `OFFENSIVE` and `HATE` classes.

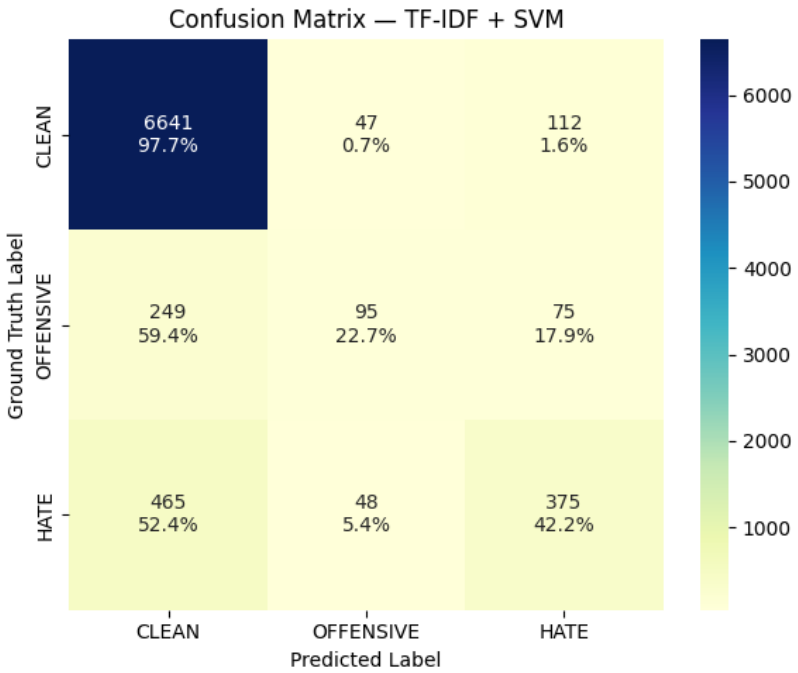
| Metric | SVM (Baseline) | PhoBERT (Transformer) |
|---|---|---|
| **Accuracy** | **87.71%** | 87.02% |
| **Macro Precision** | **0.69** | 0.65 |
| **Macro Recall** | 0.54 | **0.62** |
| **Macro F1-score** | 0.59 | **0.63** |

While the SVM model maintains a slightly higher accuracy and precision (largely by being more "conservative"and predicting the majority class more often), PhoBERT provides a much better

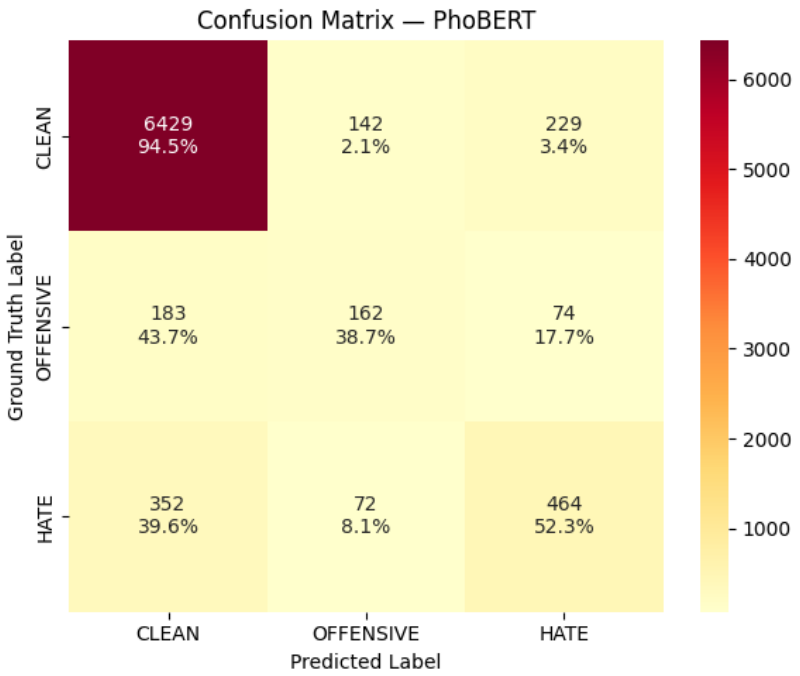
balance for toxic speech detection. Specifically, PhoBERT achieved a **Macro F1-score of 0.63**, outperforming the SVM's 0.59. This improvement is driven by a significant gain in **Macro Recall (0.62 vs 0.54)**, showing that the transformer architecture is much more effective at identifying toxic content that traditional methods miss.

### 4.3 Confusion Matrix and Class-Level Sensitivity

The confusion matrices provide a clear explanation for the performance gap:




Hình 3: Confusion Matrix: TF-IDM and SVM prediction




Hình 4: Confusion Matrix: PhoBERT prediction

The confusion matrices provide a granular breakdown of where each model succeeds and fails across the three classes.

Looking at the `CLEAN` class, both models perform strongly, with the SVM correctly classifying 97.7% of clean comments and PhoBERT following closely at 94.5%. The drop in PhoBERT's `CLEAN` precision reflects its greater sensitivity to toxic cues — it incorrectly flagged 371 clean comments as toxic compared to only 159 with the SVM. While this increase in false positives is a trade-off, it is generally considered acceptable in safety-critical NLP tasks where missing toxic content is more costly than over-flagging clean content.

Both models struggle most with the `OFFENSIVE` class, which recorded the lowest recall for either approach. The SVM correctly identified only 22.7% of offensive comments, misclassifying the majority as `CLEAN` (59.4%). PhoBERT improved this substantially to 38.7%, reducing the `OFFENSIVE` → `CLEAN` misclassification rate to 43.7%, but a large portion of offensive content still leaks through undetected. This persistent difficulty likely stems from the nature of offensive speech itself — it often relies on implicit frustration, sarcasm, or profanity-adjacent language that lacks the more explicit markers associated with hate speech, making it harder for either model to reliably distinguish from neutral commentary.

On the `HATE` class, both models perform better than on `OFFENSIVE` in relative terms, though still far from reliable. The SVM correctly identified 42.2% of hate speech instances, while PhoBERT improved this to 52.3%, reducing the `HATE` → `CLEAN` misclassification from 465 instances down to 352. The gap between the two models is most visible here, with PhoBERT's contextual understanding proving meaningfully more effective at catching content the SVM tends to dismiss as `CLEAN`.

### 4.4 Error Analysis

A manual inspection of the prediction logs reveals recurring linguistic patterns that led to misclassifications:

1. **Neutral/Clean Text Flagged as Hate:** Comments containing neutral statements or lacking malicious intent were sometimes aggressively misclassified as `HATE` (Label 2) instead of `CLEAN` (Label 0). *Example:* `"Tại vì refund có 4 người cùng nhau đạt top 1 c..."` (True: `CLEAN`, Pred: `HATE`). The model struggles with certain conversational contexts, hallucinating toxicity where none exists.

2. **Profanity/Offensiveness Missed as Clean:** Conversely, comments containing clear frustration or profanity were incorrectly passed as `CLEAN` (Label 0) rather than `OFFENSIVE` (Label 1). *Example:* `"Má..... xăng xuống giá mà luật CA cứ tăng lên"` (True: `OFFENSIVE`, Pred: `CLEAN`). The model failed to capture the `OFFENSIVE` nature of the abbreviation/profanity at the beginning of the sentence, likely confusing it with general neutral commentary on social issues.

| | dataset | type | comment | label | prediction |
|---|---|---|---|---|---|
| 12886 | ViHSD | train | Má, .....xăng xuống giá mà luật CA cứ tăng tên | 1 | 0 |
| 40019 | ViCTSD | test | sao ko lấy lao mà đâm nó , để nó tấn công vậy ... | 2 | 0 |
| 28835 | ViHSD | test | Nghe cai giong thick wa nhung tra biet ... | 0 | 2 |
| 29484 | ViHSD | test | Tôi không chạy cũng không xin, không thoái thá... | 0 | 2 |
| 16127 | ViHSD | train | Con gia nay chuyen gioi ho nhin ge qua | 2 | 0 |
| 18366 | ViHSD | train | Tại vì refund có 4 người cùng nhau đạt top 1 c... | 0 | 2 |

Hình 5: Detailed Prediction Error Log

## 5 Conclusion

This project demonstrated that while the TF-IDF + SVM baseline offers a computationally efficient and competitive solution, PhoBERT's transformer-based architecture provides meaningfully superior performance for Vietnamese `HATE` speech detection. Despite a marginal difference in overall accuracy (87.02% vs. 87.71%), PhoBERT achieved a higher Macro F1-score (0.63 vs. 0.59) and substantially better Macro Recall (0.62 vs. 0.54), confirming that it is more effective at identifying toxic content that traditional statistical methods tend to miss.

Nevertheless, both models still show considerable room for improvement, particularly in detecting offensive speech, which remained the most challenging category for either approach. Future work should focus on data augmentation strategies — such as back-translation or paraphrase generation — to better represent minority classes during training, as well as exploring class-weighted loss functions to make the model more sensitive to underrepresented categories. For those with access to greater computational resources, experimenting with `phobert-large` may also be worth exploring.